2024 Ph H2 Q14

Section: Electricity

Topic: EMF, Internal Resistance & Short-circuit Current

Summary of Question

A battery with EMF E and internal resistance r is connected to a variable external resistor R. The technician plots a graph of $1/I$ (A^{-1}) against R (Ω). From conservation of energy and $V = IR$ in the complete circuit:

$I = E/(R + r) \Rightarrow 1/I = (R + r)/E = (1/E)R + r/E$. This is a straight line with gradient $1/E$ and y-intercept r/E.

(a)(i) Internal resistance r

From the graph: y-intercept = $r/E \approx 0.20\ A^{-1}$ and gradient = $1/E \approx 0.053\ A^{-1}\ \Omega^{-1}$.

$E = 1/(\text{gradient}) \approx 1/0.053 \approx 19$ V.

$r = E \times (\text{intercept}) = 19 \times 0.20 = 3.8\ \Omega$.

Final answer: $r \approx 3.8\ \Omega$

(a)(ii) EMF E

From the gradient: gradient = $1/E \Rightarrow E = 1/(\text{gradient})$.

Final answer: $E \approx 19$ V

(a)(iii) Short-circuit current I_sc

When $R = 0$, $1/I = r/E$ = y-intercept. Therefore $I_{sc} = 1/(\text{intercept})$.

$I_{sc} = 1 / 0.20 = 5$ A.

Final answer: 5.0 A

(b) Adding a second variable resistor in parallel

Short-circuit current occurs when the external resistance is zero. Putting resistors in parallel affects the load R, but for a short circuit the only resistance is the internal r and E is unchanged.

Final answer: the short-circuit current is the same as before, because $I_{sc} = E/r$ depends only on E and r.

Revision Tips

If given a 1/I vs R graph, use: gradient = $1/E$ and intercept = r/E.

You can read I_{sc} directly as 1/(intercept) without first finding E or r.

Always quote units: E in volts, r in ohms, I_{sc} in amperes.